

NORTHWIND SALS DASHBOARD

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1. Introduction

For this project, I created a Sales Performance Dashboard using Microsoft Power BI. The dashboard is designed to analyse sales and order performance using information related to orders, products, categories, customers, employees and shippers.

The main purpose of the dashboard is to convert sales and order data into simple and meaningful visual information. It provides an overview of total orders, quantity sold, sales, freight and customers, while allowing the user to analyse performance from different business perspectives.

The interactive dashboard helps users compare sales across product categories, orders across months, employee-level order performance and customers across countries.

2. Objectives

The main objectives of this project are:

- To analyse overall sales and order performance.
- To understand total orders and quantity sold.
- To analyse total and average freight.
- To understand the number of customers.
- To compare sales across different product categories.
- To track order performance across different months.
- To compare order performance across employees.
- To analyse customers based on their countries.
- To examine product-level sales and order information.
- To create an interactive dashboard using Power BI.
- To support better business analysis through data visualization.

3. Dataset Description

The dataset used for this project contains sales and order information along with related product, category, customer, employee and shipping information. The data is organised into different tables, which are connected to analyse sales performance from different business perspectives.

The major information used in the dashboard includes:

Category Information

- Category ID
- Category Name
- Description

Customer Information

- Customer ID
- Company Name
- Contact Name
- Contact Title
- City
- Country

Employee Information

- Employee ID
- Employee Name
- Title
- City
- Country
- Reports To

Order Details

- Order ID
- Product ID
- Unit Price
- Quantity
- Discount

Order Information

- Order ID
- Customer ID
- Employee ID
- Order Date
- Required Date
- Shipped Date
- Shipper ID
- Freight

Product Information

- Product ID
- Product Name
- Quantity Per Unit
- Unit Price
- Discontinued
- Category ID
- Price Category

Shipper Information

- Shipper ID
- Company Name

These fields allow sales and order performance to be analysed across products, categories, customers, employees and shipping information.

4. Data Preparation

Before creating the dashboard, the data was organised in Power BI so that sales and order information could be analysed using the related tables.

The main preparation activities included:

- Organising order information using the Orders table.
- Using Order Details for quantity, unit price and discount information.
- Organising Products and Categories into related tables.
- Organising Customers and Employees for analysis.
- Checking important numerical fields such as Freight, Quantity and Unit Price.
- Establishing relationships between the related tables.
- Creating DAX measures for the key performance indicators.

This preparation helped create an organised data model and made the dashboard easier to analyse.

5. DAX Measures Used

DAX measures were created to calculate the important performance indicators used in the dashboard.

DAX Measure	Purpose
Total Orders	Calculates the total number of orders.
Total Freight	Calculates the total freight amount.
Average Freight	Calculates the average freight amount.
Number of Customers	Calculates the number of customers.
Total Quantity Sold	Calculates the total quantity of products sold.
Total Sales	Calculates the total sales amount using unit price, quantity and discount.

These measures help provide a quick summary of business performance and are also used in the dashboard visuals.

6. Dashboard Components

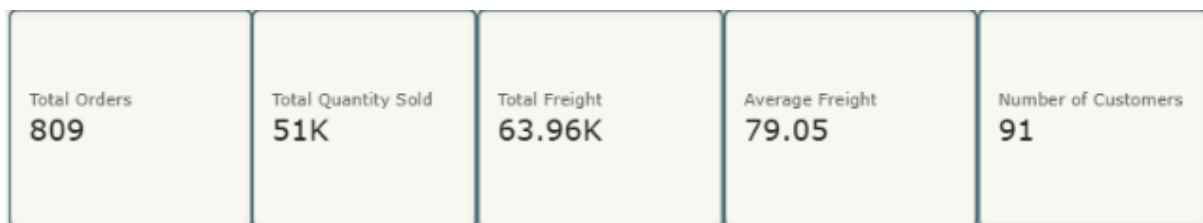
KPI Cards

The dashboard contains KPI cards that provide a quick summary of important performance indicators.

The KPIs include:

- Total Orders
- Total Quantity Sold
- Total Freight
- Average Freight
- Number of Customers

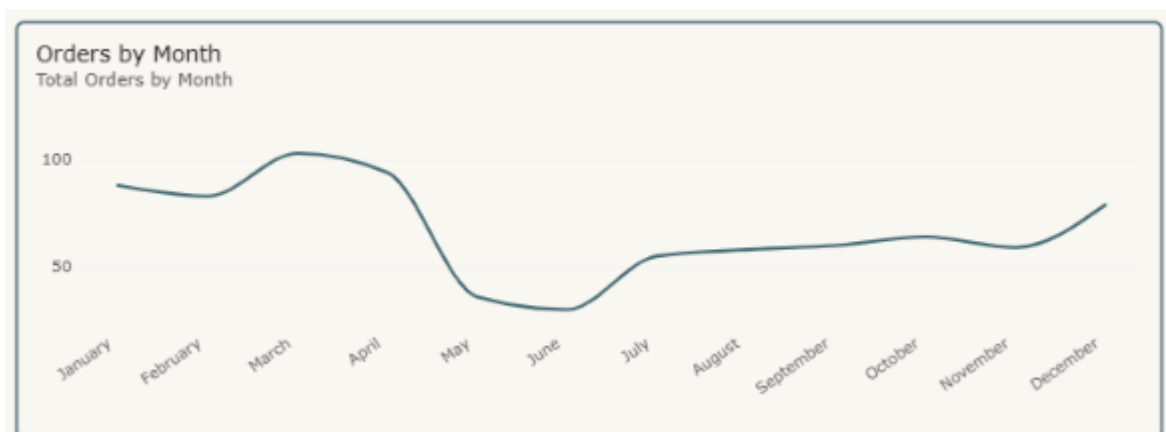
These cards help users understand the overall performance of the business at a glance.



Orders by Month

A line chart is used to display the number of orders across different months.

This visual helps users understand changes in order activity over time and identify periods with relatively higher or lower orders.



Sales by Product Category

A column chart is used to compare total sales across different product categories.

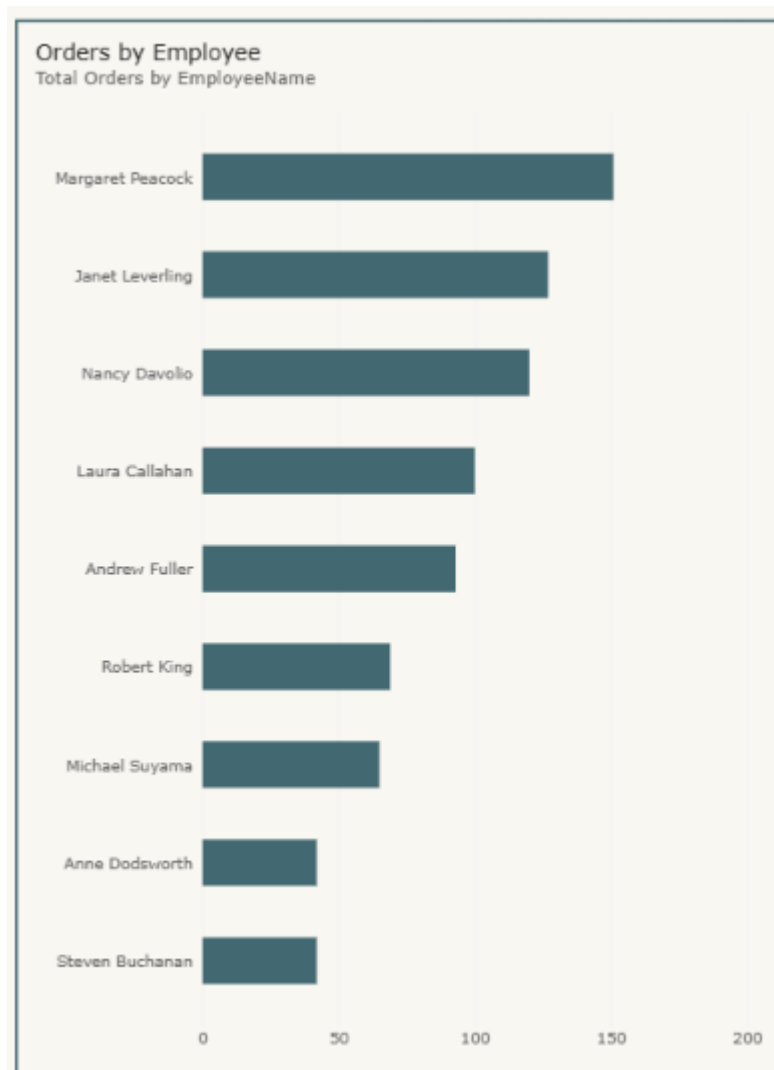
This visual helps identify which categories contribute more to total sales and which categories have comparatively lower sales.



Orders by Employee

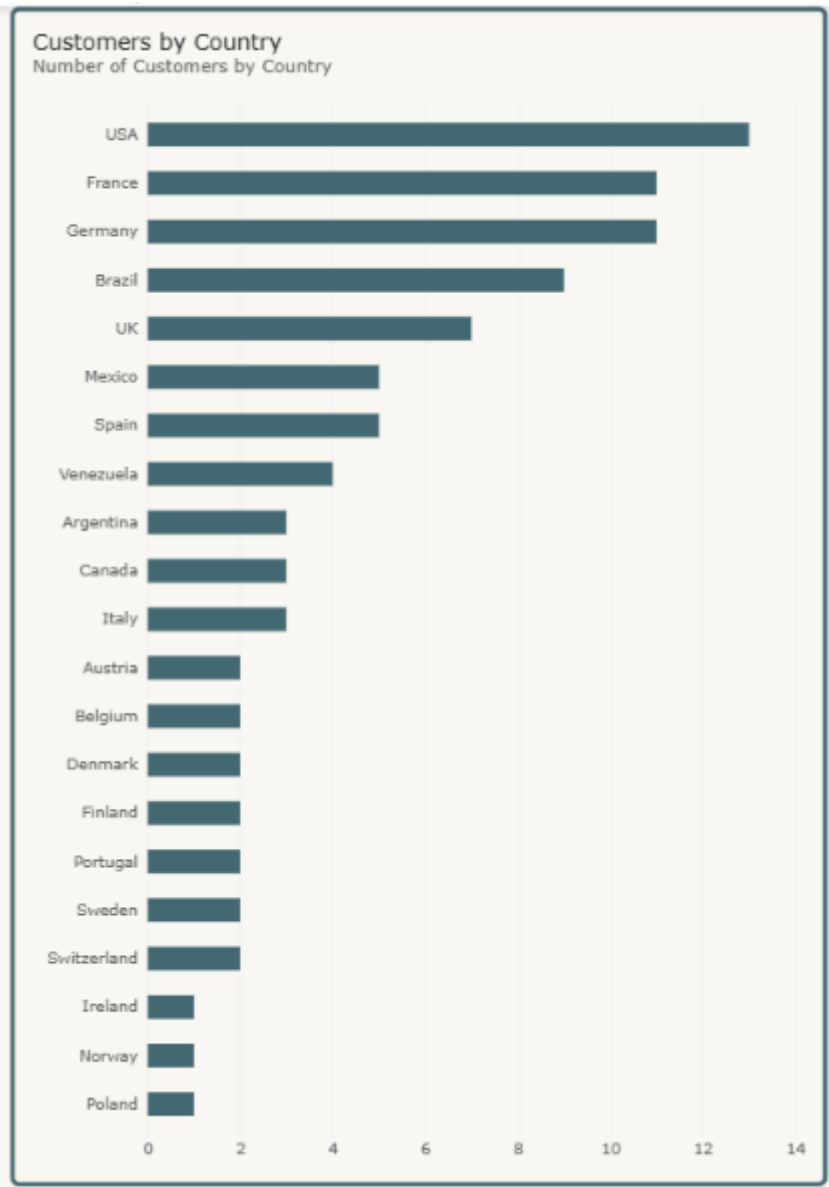
A bar chart is used to compare the number of orders handled by different employees.

This provides a view of differences in order activity among employees and helps analyse employee-level performance.



Customers by Country

A bar chart is used to display the number of customers according to different countries. This visual helps compare customer distribution across different geographical locations.



Product Details Table

A table is used to provide detailed product-level information. The table includes product, category and order-detail information. This allows users to examine individual products and their related sales information in greater detail.

ProductName	CategoryName	Total Sales	Total Quantity Sold
Alice Mutton	Meat & Poultry	32,698.38	978
Aniseed Syrup	Condiments	3,044.00	328
Boston Crab Meat	Seafood	17,910.63	1103
Camembert Pierrot	Dairy Products	46,825.48	1577
Carnarvon Tigers	Seafood	29,171.88	539
Chai	Beverages	12,788.10	828
Chang	Beverages	16,355.96	1057
Chartreuse verte	Beverages	12,294.54	793
Chef Anton's Cajun Seasoning	Condiments	8,567.90	453
Chef Anton's Gumbo Mix	Condiments	5,347.20	298
Chocolade	Confections	1,368.71	138
Côte de Blaye	Beverages	1,41,396.74	623
Escargots de Bourgogne	Seafood	5,881.68	534
Filo Mix	Grains & Cereals	3,232.95	500
Flotemysost	Dairy Products	19,551.03	1057
Geitost	Dairy Products	1,648.13	755
Genen Shouyu	Condiments	1,784.83	122
Gnocchi di nonna Alice	Grains & Cereals	42,593.06	1263
Gorgonzola Telino	Dairy Products	14,920.88	1397
Grandma's Boysenberry Spread	Condiments	7,137.00	301
Gravad lax	Seafood	2,688.40	125
Guarana Fantastica	Beverages	4,504.37	1125
Gudbrandsdalsost	Dairy Products	21,942.36	714
Gula Malacca	Condiments	9,915.95	601
Gumbär Gummibärchen	Confections	19,849.14	753
Gustaf's Knackebröd	Grains & Cereals	7,122.36	348
Ikura	Seafood	20,867.34	742
Inlagd Sill	Seafood	13,458.46	805
Ipoh Coffee	Beverages	23,526.70	580
Jack's New England Clam Chowder	Seafood	8,680.35	981
Konbu	Seafood	4,960.44	891
Lakkalikööri	Beverages	15,760.44	981
Laughing Lumberjack Lager	Beverages	2,396.80	184
Longlife Tofu	Produce	2,432.50	297
Louisiana Fiery Hot Pepper Sauce	Condiments	13,869.89	745
Louisiana Hot Spiced Okra	Condiments	3,383.00	239
Manjimup Dried Apples	Produce	41,819.65	886
Total		12,65,793.04	51317

7.Slicer Used

Three slicers are used to make the dashboard interactive:

- Year
- Country
- Category

The slicers allow users to select a particular year, customer country or product category.

When a selection is made, the dashboard visuals update accordingly. This makes it easier to analyse specific parts of the sales data instead of viewing only the overall results.

Year

2013

2014

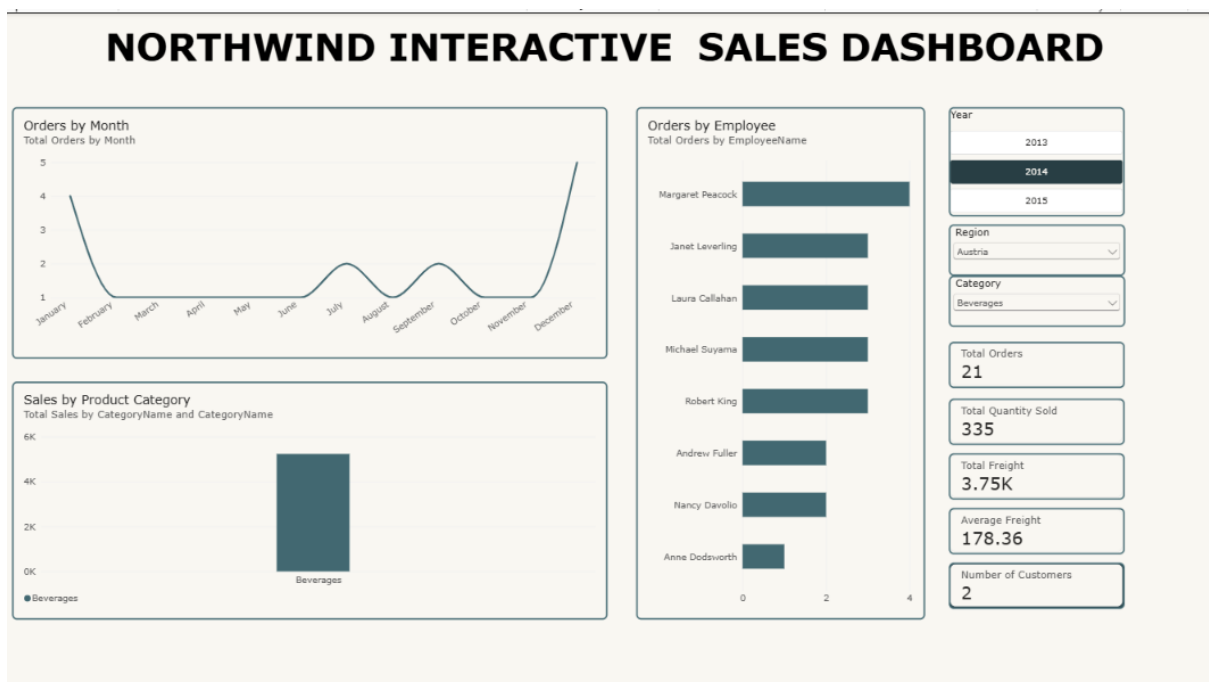
2015

Region

All

Category

All



8.Key Findings

The Sales Performance Dashboard provides the following important areas of analysis:

- Total Orders provides an overview of the number of orders.
- Total Quantity Sold helps understand the quantity of products sold.
- Total Freight shows the overall freight amount.
- Average Freight helps understand the average freight value.
- Number of Customers provides an overview of the customer base.
- Sales by Category helps compare sales performance across different product categories.
- Monthly Order Analysis helps identify changes in order activity over time.
- Employee-level Analysis provides a comparison of orders handled by employees.
- Country-wise Analysis helps understand customer distribution across countries.

- Product-level Analysis provides detailed information about products and their related sales.
- Slicers allow users to analyse the dashboard based on year, country and category.

9. Conclusion

Working on the Sales Performance Dashboard helped me understand how Power BI can be used to convert sales and order data into meaningful business information.

The dashboard combines KPI cards, charts, a product details table and slicers to provide an interactive view of sales performance. DAX measures were used to calculate important indicators such as total orders, quantity sold, freight, average freight, number of customers and sales.

The different visuals provide useful perspectives on category-wise, monthly, employee, country and product-level performance.

Overall, this project helped me understand how Power BI, data modelling, DAX and data visualization can support business analysis and better decision-making.

10. References

- Microsoft Power BI Desktop
- Global Sales Dataset
- Power BI Data Model and Dashboard
- Microsoft Power BI Documentation